

Technical Document #856: Data Engineering - Comprehensive Overview

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Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Partitioning divides data into smaller, more manageable pieces.
- Data warehouses store structured data optimized for analytical queries.
- Data lakes store raw data in its native format.